

MGT 403R – Business Analytics for Managers

Fall Quarter

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Office Hour Appointments:

See Canvas homepage for hours and signing up for appointments
(or email for appointment)

TA Office Hours: See Canvas for links and hours.

DESCRIPTION

In today's increasingly digital economy, the availability of data is widespread and business success can depend on smart, capable, and clever use of this data. Business analytics now drives ticket pricing, retail assortments, movie recommendations, digital marketing campaigns, healthcare scheduling, crime fighting, social media investments, capacity decisions and more. Today's companies generate an immense volume of transactional data, capturing trillions of bytes of information about their customers, suppliers, and operations. Millions of networked sensors are also embedded in the physical world in devices such as computers, mobile phones, smart energy meters, automobiles and industrial machines. Further, in a digitized world, consumers communicate, browse, buy, share, and search, creating extensive trails of valuable data.

Hence, analytics — the use of data with statistical and quantitative models to make better, data-driven business decisions — has become important to gain competitive advantages. With all of these different sources of data ranging from online sensors, email campaigns, to physical sensors, **business analytics is pervasive in any size business.**

This importance of analytics means that all managers should be comfortable discussing, assessing, and effectively conducting their own analyses. There is a significant shortage (*and significant demand!*) of business analysts and project managers who are adept at effectively utilizing analytics to create substantial value to an organization by solving impactful business problems.

This course introduces students to **the foundations of business analytics**. It teaches managers how to be comfortable exploring their data, how to create and use models to make better decisions in complex settings, and how to use statistics to better understand their business setting. Prior analytical experience is not a prerequisite for this course, we will introduce all of the topics throughout this course, and students of all different backgrounds have found the class useful and approachable.

OBJECTIVES

There are three main Units we cover in this course. Each Unit teaches you tools to answer key data-driven and quantitative questions. Each of the three units center around a central theme of addressing and studying **uncertainty**. Data and business analytics allows us to better understand and deal with uncertainty.

Unit 1: Uncertainty (& Randomness)

- Modeling randomness to understand what risks and potential outcomes you face.
- Understanding **why** data gives us confidence and interpreting such confidence.
- Making evidence-based business decisions (“hypothesis testing”).
 - **Is the evidence sufficient to convince us? Or is it just noise? [Key business question.]**
- Use data to create predictions for the future (“regression”).
- Use regression to interpret the relationships between data.
 - Understand the limitations of regression and how to address those limitations.
 - Understand the limitations of **causality interpretation**.

Unit 2: Decision Analysis

In Unit 2, we build on the building blocks of uncertainty that we study in Unit 1 and utilize them to structure our decision-making tools and process.

- Decisions when facing uncertain and probabilistic (“random”) future outcomes.
- Test the importance of our assumptions in our models (“sensitivity analysis”).
- Use simulation to predict complex future outcomes (such as long-tail insurance risks).

Unit 3: Optimization

In optimization, we transition from modeling random outcomes to building tools to optimize decisions, such as how many resources to allocate to different sectors or investments. Each use of resources has tradeoffs, which we learn to use tools to quantitatively model and consider.

- Simulating possible futures, and using this simulation to make the optimal decision.
- Formalizing the resource tradeoffs you face in a business into an **optimization model**.
- Using optimization model to understand the best use of many scarce resources, when faced with many tradeoffs.
- Advanced optimization models using integer and binary constraints.

All of these are key ways businesses and managers must think about their problems. This course will give you ample experience in these fundamental ways of **quantitative thinking**.

RADIANT TOOL

The central tool we use in this course to perform analysis is Radiant.

The modern use of statistics and analytics heavily relies on statistical software to conduct analyses. “R” is one of the most famous and is heavily used throughout all companies. However, using R requires being comfortable with coding first.

To best teach analytical thinking, we use Radiant to give you the power of exploring your data, visualizing it, and seeing analysis *without needing to code*. Radiant is built on-top of R, meaning it is free and it can be used at any business that can use R — which again, almost all modern analytics teams support the use of R.

Lastly and most importantly, Radiant is a valuable resume piece. This course will not only give you comfort with “data visualization”, “business analytics”, “regression analysis”, and “linear optimization” — you will also have experience with “R statistical analysis (via the Radiant interface)”.

MATERIALS

Required textbook (provided digitally, though physical versions can be ordered online)

- *Data, Models, and Decisions: The Fundamentals of Management Science*, by Dimitris Bertsimas and Robert M. Freund, 2004, Dynamic Ideas, Belmont, MA, ISBN 0-9759146-0-X.

SCHEDULE

Date	Class Topic & Activities
Week 0	Radiant Practice Video + Rady Math Bootcamp Homework (>>due in Week 1<<)
Week 1	Unit 1: Modeling Probability Distributions in Simulation
Week 2 (in-person)	Unit 1: Hypothesis Testing, Confidence Intervals with Data
Week 3	Unit 1: Introduction to Regression
Week 4	Unit 1: Regression Tricks and Trips
Week 5	Unit 2: Decision Analysis, Sensitivity Testing
Week 6 (in-person)	Unit 2: Simulation to Analyze Complex Decisions and Probabilities
Week 7	Unit 3: Optimizing Decisions in Simulation
Week 8	Unit 3: Linear Optimization Models
Week 9	Unit 3: Advances in Linear Optimization Methods
Week 10 (in-person)	Final Project Presentations + Special Topics in Analytics

DELIVERABLES

This course has a consistent workload throughout the fast-paced quarter. In particular, be wary that the beginning of the course contains a lot of material as we establish the foundation and your comfort using Radiant.

It has a substantial amount of deliverables to ensure that students have enough practice with the material, both on fundamental aspects and more in-depth analyses (*via cases*).

ASSIGNMENTS

There will be eight (8) weekly assignments in this course.

- Assignments are due via Canvas quizzes.
 - All assignment Quizzes can be taken **TWICE**.
 - This encourages the HW to be a learning tool.
 - In your first attempt, you can try your answers without concern for your grade. Then, Canvas will show you which ones you got wrong. This allows you an opportunity to figure out why it was wrong or ask for clarification, and then re-submit the quiz.
- The Canvas quiz will automatically grade your answers, but you must still submit your all of your handwritten or Radiant work to the Canvas quiz. Proof of work can be shown using screenshots of Radiant in Word documents or using Radiant HTML reports.
- The TA will check all student submissions for proof of their work.
- You are allowed to discuss homework topics at the conceptual level. If you collaborated with classmates on your homework, please note who you collaborated with at the top of your submitted homework.
- The quiz does not automatically provide partial credit. If you feel there should be partial credit for your homework, please email your course TA within **3 DAYS** of the homework due date.

CASES ASSIGNMENTS

There are 5 cases due this course in addition to weekly assignment quizzes.

- These cases consist of a slightly more in-depth analysis in a specific business and industry setting. The cases provide helpful background on a relevant business problem and background setting. The analyses for the cases will be slightly more difficult, however they can be solved using the same tools you learned in each week's class.
- Case Quiz: Each case assignment will include several quiz questions to test you on the analytical aspects of the case. Similar to the assignment quizzes, these quizzes can also be submitted twice.
- Case Writeup:
- Each case will also require you to submit a 1-2 page brief writeup on a few key aspects on each case problem. In this writeup, you will be asked to write in non-technical language suitable for a CEO, board member, or potential investor. Bullet points or short paragraphs are okay, but still, full sentences are required. Consider each case writeup as a brief executive summary.
- Specifically, you will cover:
 - (a) what are the background inputs you used for the analysis, and key assumptions,
 - (b) a high-level explanation of the kind of analysis you performed and the intuition behind it,
 - (c) discuss some high-level key takeaways from the analysis, and,
 - (d) importantly, as an open-ended discussion question, what might be some flaws in this analysis, or assumptions you may want to question.
- A detailed rubric and example of this will be provided in Week 2's lecture.
- The goal for these case writeups will be to practice with not just performing analytics in a real-world business setting, but being able to (1) communicate it in a non-technical way, and (2) consider how it could be improved.

READING // RADIANT TUTORIALS

There is required reading **before** each week (including before Week 1). In this class, I will not require you to memorize formulas. Thus, I prefer that you focus on the concepts and examples illustrated in the books, rather than the math formulas. That said, the math can often help further convey the key concept of the topic.

Preparation reading for each week:

0		Radiant data exploration and visualization practice [see Canvas for video]
1		Sections 2.1, 2.4 – 2.7, 3.1, 3.4
2		Sections 3.7, 4.1 – 4.3, 4.7, 4.8 + (Read <i>Hypothesis Testing Primer</i> after class.)
3		Section 2.9, Section 6.1 – 6.4
4		Section 6.5 – 6.10
5		Section 1 <i>and</i> the “Kendall Crab Case” in Section 1
6		Section 5.1 – 5.3, 5.7 – 5.9, <i>and</i> “Gentle Lentil Case” in Section 5 (we will discuss case in class)
7		None
8		Section 7.1 – 7.6
9		Section 9 (optional)
10		None

EXAMS

Final Project

- The final project will be a take-home project report and a short presentation in Week 10.
- You will analyze one dataset thoroughly, using many of the key tools we have learned in class. I will provide a set of existing datasets curated by me. Even better though, you can choose your own datasets that you are interested in analyzing from your work, past work, or industry domain of interest.
- You will be graded on the correctness of your application and interpretation of each topic's analysis.
- The deliverables of this take-home data project will be a report summarizing the data and analyses, a short set of a few slides introducing your data and discussing 1-2 interesting analyses, and a brief in-class presentation.
- The final project is a **great takeaway** from the class to illustrate on your resume or portfolio. It illustrates your ability to take an existing dataset and idea, and then complete all the necessary aspects of a thorough analysis to derive useful business insights.

GRADING

Exams and Deliverables	Percentage
Final Project Writeup	40%
Final Project Presentation	5%
Individual Case Quizzes and Writeups	25%
Individual Homework Assignment Quizzes	30%
Total	100%

Language Models AI + ChatGPT POLICY

ChatGPT and large-language AI models have become popular in the recent years. They can answer simple statistical questions. Modern statistical analysts and programmers can use these, however, they must also be wary that they can make simple mistakes too.

If you find it helpful, you are welcome to try to use ChatGPT to offer advice or additional support on your homework and cases. However, you must ultimately do the work on your own in Radiant and show the supporting evidence. **We expect each student to upload their own Radiant state (and supporting) files to demonstrate their ability to perform the analysis themselves without AI support.** And for the case writing, you are expected to write any discussions and include your own Radiant analysis. The TA will be particularly vigilant in their grading of any non-sequitur output that a ChatGPT-type model might output.

Furthermore, for the take-home project, you must write the report and answer the analytics questions on your own. Given that this is an MBA course, the goal is primarily your education and growth, not a particular letter grade. The final has been one of the best learning resources for students in the past, so I encourage you to write and analyze the take-home project primarily on your own. The final exam also requires **uploading your own Radiant state file** as supporting evidence.

COURSE SUPPORT

Business Analytics/Quant can be a difficult first-course at Rady, especially if you fall behind on a week. It can also be difficult in the first few weeks as you get comfortable with Radiant and the foundations of the class. We recognize this and have set up many avenues for support for this class

TA Office Hours

- We will seek to provide variety in the available TA office hours via Zoom. These will be posted on the Canvas homepage. TA office hours can be used for technical help and understanding class concepts or problems.
- They will be on Zoom and have multiple “break-out rooms”. You can join with your study group, or other students there to allow them to be co-operative **work sessions**. In each breakout room, you can “*raise your hand*” on Zoom to request a TA to step in. The TA’s will be asked also to occasionally pop into your breakout rooms to see if they can help.

Canvas Discussions

- In the course Canvas, there is a Discussions section on the sidebar where students can ask questions and other students can help answer these questions to help and test your learning.
- Both I and a TA will work to provide answers quickly to clarify and answer the questions.

Professor Office Hours

- I will be available for 1:1 office hours once a week. The Canvas homepage will have the up-to-date hours, and 20-minute time slots can be reserved ahead of time. Please reserve them at least a day in advance, otherwise I may be unavailable for last-minute bookings. These will be on Zoom.
- See Prof. Zhe office hour details and reservations system on the Canvas homepage.

Extensions

- If difficult situations come up for you, please do not hesitate to email me and we can discuss ways to adjust homework deadlines.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:

<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.